

MICROMOLD · DESIGN PRECIS

MicroMold design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ d0d37e2		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 update from MMD-CAL-001 and MMD-DDR-001 (ratchet handle, 4 mm nozzle, load cell spacer, 20 mm bracket, nozzle zone shield, side fume hood, raised nozzle; numbers and design choices updated)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, components, first-order numbers, design choices, safety, open questions, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

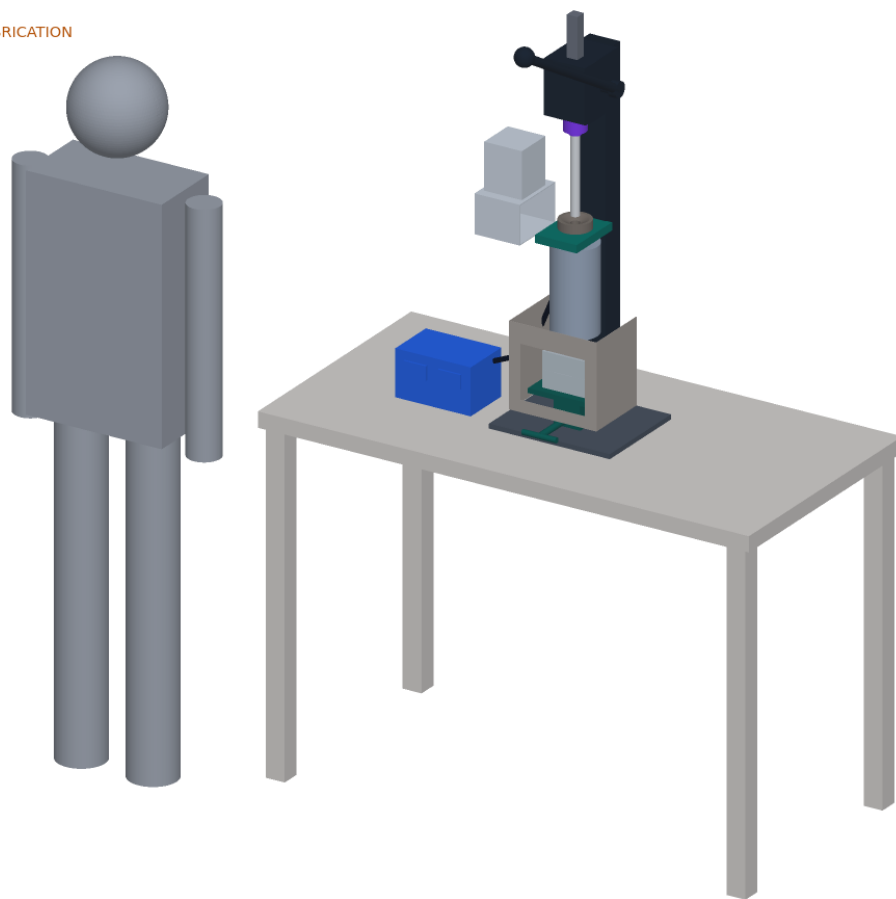
MicroMold design precis

■ Summary

MicroMold is a bench-top, hand-operated plunger injection press for recycled HDPE, PP, LDPE and PS. The rack-and-pinion head of a 1 t arbor press, mounted on a taller steel column and turned by a 450 mm ratchet handle, drives a 22 mm plunger down a vertical steel barrel heated by two 250 W band heaters. Melt leaves a heated 4 mm nozzle into a two-plate aluminum mold that a screw lift table holds against the nozzle, inside a perforated nozzle zone shield, and a side hood with a duct fan draws fumes from the funnel. The TRL 3 calculations (MMD-CAL-001) give a shot of up to 34.2 g at 8.9 MPa (89 bar) with 250 N on the handle, delivered in about four ratchet pulls; a warm-up of 14.8 min; and about 9.4 parts per hour from 635 W of single-phase power. The press costs \$495 in parts and \$585 with one mold, over the \$400 budget, and weighs about 38.5 kg, over the 35 kg target. All figures are estimates.

MicroMold

CONCEPT, NOT FOR FABRICATION



Grey: workbench (0.9 m), 1.75 m person for scale

Figure 1. MicroMold on a 0.9 m workbench with a 1.75 m person for scale. Plunger raised for loading; test mold seated on the nozzle inside the shield. Generated from the parametric model `cad/src/model.py`.

■ How it works

1. **Heat.** Two PID controllers drive the barrel zone (two 250 W band heaters) and the nozzle zone (a 100 W band heater) to the set point for the resin, for example 210 °C for HDPE. An insulation jacket and a perforated guard keep the outer surface near 48 °C. The duct fan runs whenever the heaters are on.
2. **Load.** With the plunger raised, the operator pours washed, dried flake into the funnel and tamps each of two or three top-ups down with the press itself, because loose flake has about a third to a half of the density of melt and an untamped charge wastes stroke (MMD-CAL-001, A4).
3. **Soak.** The fresh charge melts above the melt left from the last shot. The barrel holds 2.2 shots, so each charge soaks for about two cycles, about 12 min, against the 9.3 min a 22 mm column of tamped flake needs.
4. **Clamp.** The operator bolts the two mold plates together, sets the mold on the lift table, closes the shield's front and turns the screw until the sprue bushing seats on the nozzle.
5. **Inject and hold.** The operator fills the mold in about four pulls of the ratchet handle, each moving the plunger about 31 mm, over about 10 s; the pawl holds the ram between pulls. The operator then holds pressure for about 30 s while the gate freezes. A load cell above the plunger shows its force, so the melt pressure (force divided by 380 mm² of plunger area) can be read and logged.
6. **Cool and eject.** The plaque cools in the mold for about 2 min, then the mold is lowered, unbolted and opened. Sprue and purge go back to the regrind bin.

MicroMold: material flow per shot, grams (estimates, 30 g HDPE test part)

CONCEPT, NOT FOR FABRICATION

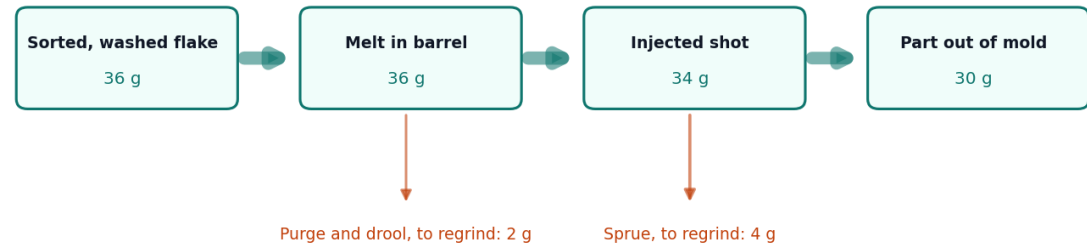


Figure 2. Material flow per shot for the 30 g reference part. All values are estimates: about 2 g of purge and drool and 4 g of sprue return to regrind.

■ Main components

Table 1. Main components. Numbers match `bom/bom.csv` and Figure 4.

#	Component	Choice	Notes
1	Base plate	320 x 260 x 12 mm steel, bolted to the bench	Carries the column and the mold clamp
2	Column and drive head	Rack-and-pinion head of a 1 t arbor press on an 80 x 60 x 3 mm steel tube column 0.9 m tall; 450 mm, 1/2 in drive ratchet handle on the pinion shaft	Ratchet added at TRL 3: one pull moves the plunger only about 31 mm
3	Rack ram	From the arbor press, 28 mm square, 255 mm or longer	150 mm of travel with the rack still engaged; to confirm on the chosen press
4	Plunger load cell	10 kN compression cell with HX711 amplifier and display, on a 10 mm G-11 glass-epoxy spacer	The spacer keeps the cell near 27 °C instead of about 94 °C
5	Plunger	22 mm ground steel rod, 200 mm, cross-drilled for a pinned floating coupling	0.05 to 0.10 mm diametral clearance (22 H8/f7); no seal
6	Heated barrel	42 mm steel bar bored and honed to 22 mm, 260 mm long, 80 mm flange and loading funnel	Machine shop part
7	Band heaters	Two 250 W mica band heaters, 42 mm ID x 50 mm	Barrel zone, one PID loop; 3.8 W/cm ²
8	Nozzle and nozzle heater	Steel nozzle with a 4 mm orifice, radiused tip; 100 W band heater; tip at 190 mm above the bench	Second PID loop; orifice enlarged from 3 mm at TRL 3
9	Barrel bracket and heat break	20 mm steel plate welded to the column; the flange sits on four mica pads	Carries the plunger reaction into the column; thickened from 10 mm
10	Insulation jacket and guard	25 mm mineral wool in an aluminum skin, perforated steel guard	Skin about 48 °C
11	Mold clamp	Tr20 x 4 screw jack with handwheel, 170 x 130 mm lift table, 100 mm travel	Table top 60 to 160 mm: stacks of 30 to 120 mm
12	Aluminum mold set	Two 6061 plates, 120 x 90 x 45 mm each, four M10 x 110 bolts, sprue tapering 5 to 7 mm	First mold: test plaque; tooling outside the press budget
13	Control box	Two PID controllers with set point limit, two 25 A SSRs on a heat sink, fused IEC inlet, double-pole switch, independent thermal cut-out, fan outlet, force display	Mains enclosure; earthed
14	Heater and sensor wiring	High-temperature wire in glass-fiber sleeving, two K-type thermocouples	250 °C rated at the barrel
16	Nozzle zone shield	Perforated steel sides and hinged front around the nozzle and mold	Added at TRL 3 for R11 and melt spit
17	Fume hood and duct fan	Side hood, 150 x 100 mm face, 100 mm from the funnel axis; 100 mm inline fan; 3 m of duct	Added at TRL 3 (MMD-DDR-001 D9); about 124 m ³ /h

Item 15 (hardware and consumables) is in the BOM but not modeled. The general arrangement is drawing MMD-DWG-001 ([cad/drawings/](#)).

MicroMold: cutaway

CONCEPT, NOT FOR FABRICATION

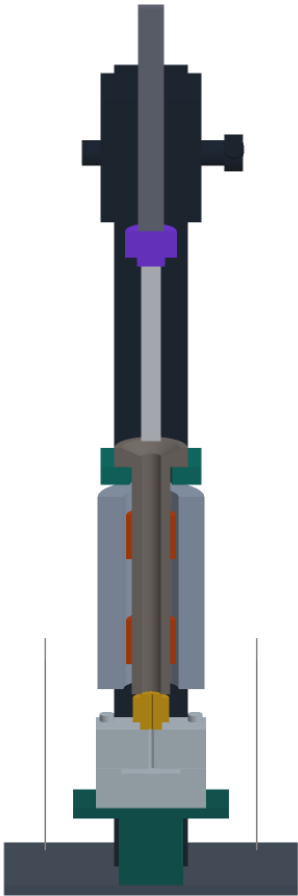


Figure 3. Section on the injection axis: drive head and ram at the top, load cell and spacer (purple), plunger raised above the funnel, barrel with the two band heaters inside the jacket and guard, bracket (teal), nozzle (gold) seated on the mold with its sprue and plaque cavity, lift table and screw jack on the base plate, and the sides of the nozzle zone shield.

■ **Numbers from MMD-CAL-001**

All values are first-principles estimates. Assumptions are in MMD-CAL-001, Table 1; the tags point to the lines of `docs/04-calcs/sizing.py` that print them.

Table 2. Key numbers.

Quantity	Value	Requirement
Swept volume and shot	45.6 cm ³ , 34.2 g HDPE ideal; about 25 g if the fresh charge is not tamped [A1, A2, A4]	R2 at risk
Drive ratio and efficiency	22.5:1; 0.60 design, 0.84 calculated [B1]	
Plunger force and melt pressure at 250 N	3.38 kN and 8.9 MPa (89 bar, 1,288 psi) at 0.60; 12.4 MPa at 0.84 [B2]	R3 met on paper, 11 % margin
Stroke per ratchet pull	31.4 mm per 90° pull; 3.8 pulls per shot [B5]	
Overload (700 N on the handle)	13.2 kN, 34.7 MPa, 1.35 times the press rating [B4]	Parts sized for it
Fill pressure, 4 mm nozzle, 10 s fill	About 4.8 MPa [E3]	
Mold clamp limit	45 cm ² projected area at 8.9 MPa [D2]	R7 met on paper
Warm-up to 220 °C	14.8 min barrel zone, 6.6 min nozzle [G4]	R5 at risk
Standing losses at 220 °C	103 W; jacket skin 48 °C [G2, G3]	R11 met on paper
Melt soak for a 22 mm column	9.3 min for tamped flake [H1]	
Cycle	6.4 min, 9.4 parts per hour in still air; mold about 92 °C [H2, H4]	R6 at risk
Electrical load	635 W; 2.8 A at 230 V, 5.3 A at 120 V [I1]	R10 met on paper
Energy per shot	About 21 Wh, 0.70 kWh per kg of parts [I2]	
Fume hood flow	About 124 m ³ /h [J1]	R13 met on paper
Mass	38.5 kg without the control box [K1]	R9 not met
Size	320 x 260 mm base; handle 1,089 mm above the bench at the start of a pull [K2]	R9 met on size
Parts cost	\$495 press; \$585 with one mold [L1]	R14 not met

■ Key design choices

These follow MMD-DDR-001. Each is adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction and is open for his review.

- **Drive: rack and pinion from an arbor press (D1).** The rack, pinion and head of a 1 t arbor press on a taller column give 22.5:1 from one bought part. A screw press stays as a variant for small, thick parts. At TRL 3 the fixed handle is replaced by a ratchet handle, because a shot needs 344° of pinion rotation.
- **Vertical barrel, mold below (D2).** Gravity keeps the melt at the nozzle and the operator loads from the top. The mold is changed at bench height.
- **Plunger, not screw (D3).** A reciprocating screw would mix and melt better but needs a motor, gearbox and a much more complex barrel.
- **Two heat zones (D4).** A separate nozzle zone stops the nozzle freezing between shots at the cost of a second PID and SSR.

- **Bolted two-plate aluminum molds on a lift table (D5).** Bolts are slow but cheap and hold 45 cm² at full pressure. A toggle clamp frame (about \$40) is a later upgrade. The nozzle tip is now 190 mm above the bench so that 120 mm stacks fit.
- **Load cell for pressure indication (D6).** A number for each shot helps a group learn and record settings. A glass-epoxy spacer keeps the cell cool.
- **First mold: a test plaque (D7).** A 64 x 50 x 6 mm plaque shows fill, shrinkage and surface quality and can be cut into test bars. Product molds follow co-design.
- **Budget (D8).** The \$400 budget covers the press; molds are tooling, reported separately (R8). The press is \$495 at TRL 3, so R14 is not met; further options are in `docs/REVIEW.md`.
- **Fume control (D9).** A side hood with a duct fan at the funnel, plus a written condition to run only under it or outdoors.
- **Co-design partner type (D10).** The first partner is to be a group that already shreds HDPE or PP; the specific partner and place remain open.

MicroMold: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Base plate
- 2 Column, drive head and ratchet handle
- 3 Rack ram
- 4 Plunger load cell and spacer
- 5 Plunger, 22 mm
- 6 Heated barrel
- 7 Band heaters, 2 x 250 W
- 8 Nozzle and nozzle heater
- 9 Barrel bracket and heat break
- 10 Insulation jacket and guard
- 11 Mold clamp (screw lift table)
- 12 Aluminum mold set
- 13 Control box with PID and SSR
- 14 Heater and sensor wiring
- 16 Nozzle zone shield
- 17 Fume hood and duct

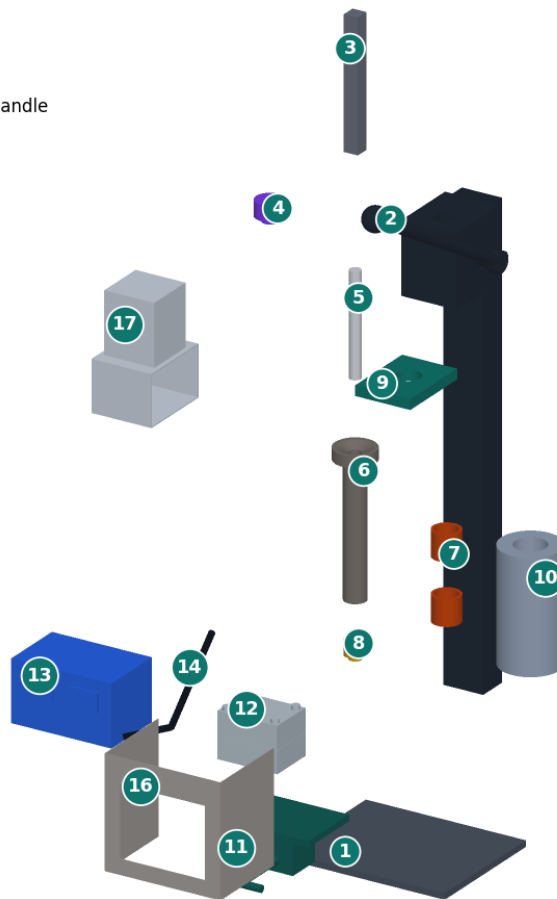


Figure 4. Exploded view with BOM numbers.

■ Safety

Safety: MicroMold combines mains voltage, surfaces and molten plastic at up to 260 °C, stored pressure in the melt, a long ratchet lever and plastic fumes. It is a concept for a supervised workshop, not a consumer appliance.

- **Hot surfaces and melt.** The barrel, nozzle, mold and purge reach 180 to 260 °C, and molten plastic sticks to skin. The funnel top is a working surface at about 150 °C. Keep the jacket guard and the nozzle zone shield in place, wear heat-resistant gloves, long sleeves and eye protection, and never look down the barrel.
- **Pressure.** Trapped melt can spit from the nozzle or the parting line. An operator hanging body weight on the handle can reach about 35 MPa, which would open the parting line of the test mold (MMD-CAL-001, D3). Close the shield before injecting, keep faces away from the axis and fit a handle stop or a pull limit.
- **Fumes.** Process only HDPE, PP, LDPE and PS within their published ranges. Never heat PVC, which releases hydrogen chloride, or unknown and mixed plastics ([Precious Plastic Academy](#)). Overheated PS can release styrene. Run the hood fan whenever the heaters are on, or work outdoors. Set point limits of 260 °C are set in both controllers.
- **Mains electricity.** Heaters run at mains voltage next to steel parts. Earth the frame and every metal part, use a fused inlet, double-pole switch and an RCD (30 mA) or GFCI supply, rate wiring at the barrel for 250 °C, and fit an independent thermal cut-out. Mains wiring must be done or checked by a qualified electrician to local electrical code.
- **Moving parts and lever.** The ratchet holds the ram under load: release the pawl only with the handle in hand. The ram and pinion can pinch; keep hands clear of the rack.
- **Fire.** Unattended heaters are a fire risk. Switch off at the inlet after use; keep flake and rags away from the barrel.
- **Product use.** Recycled parts of unknown history are not suitable for food contact, toys for young children, medical or safety-critical uses.

■ Open questions after TRL 3

- Ram length and travel of real 1 t arbor presses: is a 255 mm ram with 150 mm of engaged travel available, and how heavy is the head? This sets R9.
- How dense is a tamped, part-melted charge at injection time? This sets R2.
- Does the mold warp parts at about 92 °C in still air, and is a cooling fan enough? This sets R6.
- Nozzle temperature for each resin, and mold design rules for the group's own machinists: draft, venting, gate size and ejection.
- Which first products would sell locally? This needs the co-design partner (MMD-DDR-001, O1 and O2).

Concept media: [blueprint sheet](#), [interactive 3D model](#). General arrangement: [MMD-DWG-001](#).